

DPA 4099 ON THE YAMAHA C3

A deep dive on your usual case — 6 ft 1 in grand, stereo 4099P, open lid & closed lid

Mics: 2 × 4099 CORE+ **Extreme SPL** (yellow band) in the 4099P piano kit
Why this is the *best* home for the Extreme SPL: a grand exceeds 130 dB near the hammers.

Live Sound KB · companion to article mic-dpa-4099 (C3 section)

The C3, and why your Extreme SPL pair belongs in it

The Yamaha C3 is a 6 ft 1 in (186 cm) conservatory grand — 88 keys from A0 (27.5 Hz) to C8 (4186 Hz), duplex-scaled, with the bass strings crossing over the tenor break and a short, dense treble section. Two soundholes sit in the plate. Up close it is enormous dynamically: DPA measures more than **130 dB peak near the hammers**.

This is the best room for your Extreme SPL mics

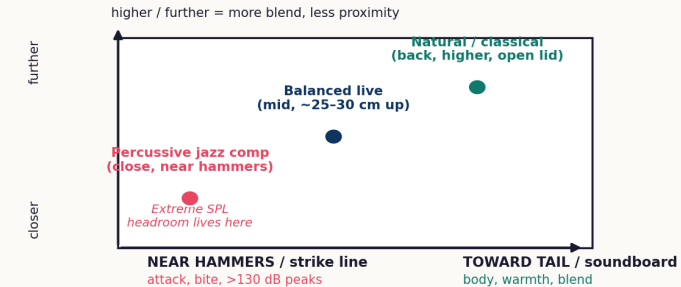
On a quiet solo violin the Extreme SPL's 28 dB(A) floor shows. Inside a C3 it's the opposite story — the instrument is loud right where the capsules sit, so the noise floor disappears and the **152 dB headroom gets used** on every hard fortissimo strike. Close, directional, loud source = exactly what the Extreme SPL was built for. Your usual gig is the ideal gig for these mics.

Close + directional = percussive + isolated

Clip mics inside the piano capture mostly direct sound and very little room, which reads as more percussive and attack-forward than a pair out in front. That's a feature on a band stage (cuts through, resists feedback) and something to soften with placement and a room blend when you want natural. The supercardioid pattern is what buys you the isolation in both your lid positions.

Where you aim sets the tone

Distance from hammers and height over strings

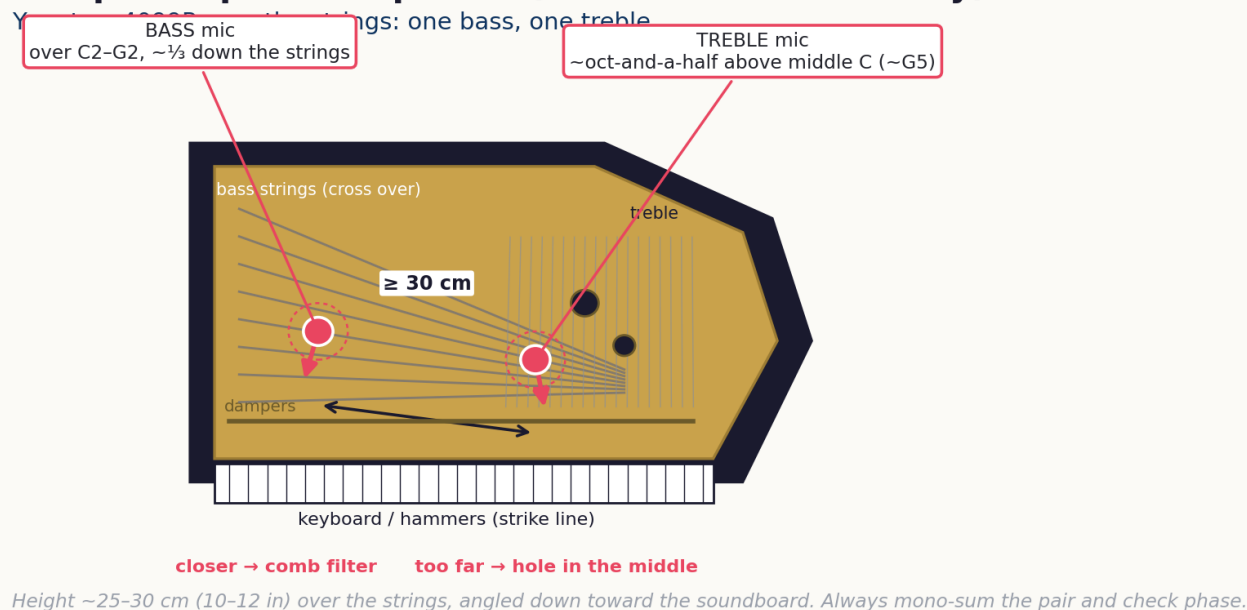


Closer + nearer the hammers = attack and proximity bass (HPF harder). Back and higher = body and room blend.

Closer and nearer the hammers buys attack and proximity bass; back and higher buys body and blend. Pick your spot on this map before you touch EQ.

Your stereo pair — where the two mics go

C3 spaced pair — top view (lid removed for clarity)



Step	Bass (low) mic	Treble (high) mic
Position	Over C2–G2, ~ $\frac{1}{3}$ of the way down the strings, well into the casing	~An octave-and-a-half above middle C (~G5)
Height	25–30 cm (10–12 in) over the strings, angled down	25–30 cm over the strings, angled down
Mount	4099P magnet to the plate; shape with the gooseneck	Same — magnet on the plate
Spacing	30 cm+ (12 in) between the two — closer combs, too far leaves a hole in the middle	—

Set-up order that works

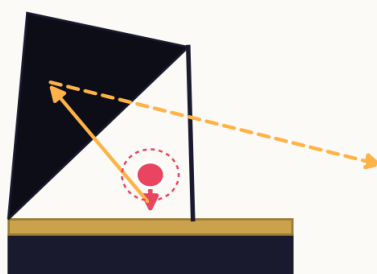
Mount both, HPF in, then bring the mics up **one at a time** with EQ flat. Mono-sum the pair and listen — if the center thins, flip polarity on one. Get a balance you like, then leave it alone. The real-world rule from engineers who live on these: once it sounds like the piano (just louder), stop touching it.

Your two lids — open and closed

Your two cases — open lid vs closed lid (side view)

Same spaced pair; the lid changes everything

FULL STICK / OPEN



natural & open · max projection
most bleed + feedback risk

CLOSED / LID DOWN



max isolation · darker & boxier
comb-filter risk · phase-check

Closed lid traps reflections under the lid → boxiness (200–400 Hz) and comb filtering. Wider mic spacing + EQ tame it.

	Full stick / OPEN	Closed / LID DOWN
Tone	Most natural and open; lid reflects to the room	Darker, more boxed-in; reflections trapped under the lid
Use it for	Solo, classical, vocal accompaniment, quieter stages	Loud bands, pit, max separation, monitors blasting
Watch for	Bleed and feedback — most of both	Boxiness (200–400 Hz) and comb filtering
Placement tweak	Mics a touch lower/closer for gain-before-feedback; let them catch a little lid reflection for air	Low-profile on the plate; widen the pair spacing to fight combing
Feedback move	Aim the supercardioid null at the nearest wedge/PA	Lid itself is the isolation; null still helps with spill

EQ, recording & feedback

Start flat. A real engineer micing a grand with 4099Ps on open/short stick left the desk essentially flat except a low-end roll-off (~90 Hz on the bass mic, a little higher on the treble). Cut only what the room shows you. The mic's built-in +2 dB at 10–12 kHz means you almost never add air. Whole-dB values.

Move	Open lid	Closed lid
HPF — bass mic	90 Hz	90 Hz
HPF — treble mic	~110 Hz	~110 Hz
Boxiness (lid)	none, or gentle -2 @ 300 Q1.4	-3 to -4 @ 250–400 Q1.5 (expect this)
Low-mid tub	-2 @ 200 Q1.5 only if boomy	-3 @ 200 Q1.5
Honk	-2 @ 500 Q1.5 if present	-3 @ 500 Q1.5
Air (10–12 kHz)	leave it — it's built in	leave it; if closed-lid-dull, gentle +2 @ 8 kHz shelf

Recording & post

The 4099 pair is dry and internal by design. For your multitrack/post work, capture each mic to its own track and blend a room layer underneath — a spaced room pair, or at the Memo your crowd rig — to put the instrument back in a space. The dry close pair gives you control; the room gives you air.

Feedback & bleed

Supercardioid rejects best at its rear/sides — point that null at the nearest monitor or PA stack. HPF clears proximity rumble and stage thumps. Closed lid is your biggest single isolation win on a loud stage; open lid leans on placement and the null instead.

Real-world reference photos

This page fills itself from the drop folder. Add photos and ask me to rebuild — see below.

No reference photos dropped yet. Save image files — DPA pages, Sound On Sound, your own C3 rig shots — into **Live Sound KB/_refs/dpa-4099-c3/**, then ask me to rebuild and they embed here, cropped and captioned. Optional captions go in captions.tsv (filename <tab> caption <tab> credit); prefix files 01_, 02_ to set order.

C3 quick-rig & real-world references

1	Magnets to the plate. Bass mic over C2–G2 (~1/3 down the strings); treble mic ~G5.
2	Both 25–30 cm above strings, angled down toward the soundboard. Pair 30 cm+ apart.
3	HPF in — adapter 80 Hz plus console ~90 Hz bass / ~110 Hz treble.
4	Mono-sum, phase-check, flip one if the center thins.
5	Bring up one mic at a time, EQ flat. Cut only what's boxy/honky.
6	Closed lid: widen the spacing, expect to pull 250–400 Hz. Open lid: mind feedback, aim the null.
7	Set & forget. Light level rides only. Record each mic to its own track + a room layer.

Getting real photos into this doc

Drop image files into **Live Sound KB/_refs/dpa-4099-c3/** (DPA, Sound On Sound, your own rig shots) and ask me to rebuild — they embed on the Real-World Reference Photos page, cropped and captioned. The diagrams in this doc are drawn faithfully from the real references; the links below go straight to the genuine photos and DPA's how-to videos.

Real-world reference gallery (click through)

Reference	What it shows	Link
DPA — How to mic a piano	Grand/upright methods + 4099 piano video; the stereo set-up diagram these drawings follow	dpamicrophones.com/mic-university/how-to-mic/how-to-mic-a-piano
DPA — 4099 piano mounting video	4099P magnet mounts seated in the frame, real piano	youtu.be/k19mByCajDo
DPA — 4099 stereo piano video	Most popular ways to mic a piano with a 4099 pair	youtu.be/otj4M3j_pWo
Sound On Sound — Miking a piano concert with DPA	Real gig: 4099Ps on a grand, magnet on the gold frame, in-situ placement photos	soundonsound.com/techniques/miking-piano-concert-dpa
SOS — How loud is a concert grand?	Backs the >130 dB-near-the-hammers figure	soundonsound.com/sound-advice/q-how-loud-concert-grand-piano
DPK4099 CORE+ piano kit	The piano stereo kit page (clips, mounts, specs)	dpamicrophones.com/microphones/kits/dpk4099